

SUPERIOR UNIVERSITY

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Section: 3A-BSSE Semester: 3rd

Professor: Muhammad Ahmed

Assignment NO: 01

Name :- Muhammad Asim
Roll No:- Su92-BSSEM-F25-054
Section :- 3A-BSSE Semester :- 3rd
Subject :- OOP Theory
Professor :- Muhammad Ahmed
Assignment No :- 01

QNo :- 01

Ans :-

```
#include<iostream>
using namespace std;

class circle
{
  private:
    float radius;
    float x;
    float y;
  public:
    circle()
    {
      radius = 0;
      x=0;
      y=0;
    }
    circle(float a , float b, float r)
    {
      x=a;
      y= b;
     if (r>=0){
        radius=r; }
     else {
       cout<<"Invalid radius!"<<endl;
        radius=0;
      }
    }
```

```
void setvalues(float a, float b, float r)
{
  x=a;
  y=b;
 if(r>=0)
   radius=r;
 else {
 cout<<"Invalid radius!"<<endl;
   radius=0;
  }
}
 float area()
 {
  return 3.1415*radius*radius;
 }
 float circumference()
 { return 2*3.1415*radius;
 }
 void display()
{
 cout<<"x="<<x<<endl;
 cout<<"y="<<y<<endl;
 cout<<"radius="<<radius<<endl;
 cout<<"Area="<<area()<<endl;
 cout<<"circumference="<<circumference()<<endl;
}
};
 int main()
{
 circle c1;
  circle c2(5,10,7);
 cout<<"circle 1:"<<endl;
  c1.display();
 cout<<"\n circle 2:"<<endl;
  c2.display();
 cout<<"\n After setting value:"<<endl;
  c1.setvalues(2,3,4);
  c1.print();
  return 0;
}
```

Qno:- 02

Ans:-

```
#include<iostream>
#include<iomanip>
using namespace std;
class Time
{
private:
  int hours;
  int minutes;
  int seconds;
Public:
  Time()
{
  hours=0;
  minutes=0;
  seconds=0;
}
Time (int h, int m, int s)
{ hours=h;
  minutes=m;
  second=s;
}
void display() const
{ cout<<setfill('0')<<setw(2)<<hours<<":"
  <<setw(2)<<minutes<<":"<<setw(2)<<seconds<<endl;
}
void add(const Time& t1, const Time& t2)
{ seconds=t1.seconds+t2.seconds;
  minutes=t1.minutes+t2.minutes;
  hours=t1.hours+t2.hours;
  If (seconds >=60)
  { seconds=seconds-60;
    minutes++;
  }
  If (minutes >=60)
  { minutes=minutes-60;
    hours++;
  }
  }
};
```

```
Int main ()
{
  Time t1(10, 45, 30);
  Time t2(5, 30, 45);
  Time t3;
Cout << "Time 1: ";
 t1.display();
Cout << "Time 2:";
 t2.display();
 t3.add(t1,t2);
Cout << "Result: ";
 t3.display();
 return 0;
}
```

QNo:- 03

```
#include<iostream>
using namespace std;
class TollBooth
{ private:
 unsigned int totalcars;
  double totalcash;
 Public:
  TollBooth()
 {
   totalcars=0;
   totalcash=0;
 }
 void payincar()
 {
   totalcars++;
   totalcash+=0.50;
 }
  void nopaycar()
  {
   totalcars++;
  }
```

```
void display() const
{
 Cout << "\n Total Cars = " << totalcars << endl;
 Cout << "Total Cash = " << totalcash << endl;
}
};
  int main()
  {
   TollBooth booth;
   char choice;
  Cout << "P= Paying Car " << endl;
  Cout << "N= Non-Paying Car " << endl;
  Cout << "E=Exit" << endl;
  while (true)
  {
  Cout << "\n Enter choice: ";
   cin >> choice;
   if (choice == 'P' || choice == 'P')
   {
    booth.payingcar();
   }
   else if (choice == 'N' || choice == 'n')
   {
    booth.nopaycar();
   }
   else if (choice == 'E' || choice == 'e')
   {
    booth.display();
      break;
   }
   else {
    Cout << "Invalid choice." << endl;
   }
  }
  return 0;
}
```

Q no:- 04

```
#include<iostream>
using namespace std;

Class Angle
{
private:
  int degrees;
  double minutes;
  char direction;
Public:
  Agle ()
  { degrees = 0;
    minutes = 0;
    direction = 'N';
  }
  Angle (int d, double m, char dir)
  { degrees = d;
    minutes = m;
    direction = dir;
  }
  Void getAngle()
  { Cout<<"Enter degrees: ";
    cin>>degrees;
    Cout<<"Enter minutes: ";
    cin >> minutes;
    Cout<< "Enter direction: ";
    cin>> direction;
  }
  void display() Const
{
```

```
    cout << degress << "°" << minutes << "'" << direction << endl;
  }
};

int main()
{
  Angle latitude;
  Angle longitude;
  cout<< "Enter latitude" << endl;
   latitude.getAngle();
  cout<< "\nEnter longitude:" << endl;
   longitude.getAngle();
  cout<< "\n latitude:";
   latitude.display();
   cout<< "Longitude: ";
   longitude.display();
   return 0;

  }
```

Q No :- 05

Critical Inquiry-based Challenge

A. Problem Analysis

1- Wrong input.

The user may enter wrong value, such as a negative radius.

Cause :- No input checking.

Effect :- The program may give a wrong result.

Solution :- Check the input before saving it.

2- Wrong Time

The user may enter 75 minutes or 90 seconds.

Effect :- The time result may be wrong.

Cause :- Time value are not checked.

Solution :- Check and correct minutes and seconds.

3 :- Data can change

Object data may be changed by mistake

Cause :- Const is not used

Effect :- Data may become incorrect

Solution :- Use Const for function that only read data.

B. Inquiry and Exploration

1. wrong input :

1. Check the input.
2. Ask the user to enter the value again if it is wrong.

2- Wrong Time:.

-1 Check that minutes and second are between 0 and 59.

-2 Convert 60 seconds into minutes and 60 minute into 1 hours.

3- Data Change

-1 Use Const for read-only functions

-2 Use Const object when their values should not change.

C. OOP Design:-

Circle class:-

Data: radius, x, y

Function:- area(), circumference(), print()

validation: Radius cannot be negative.

Time class

Data:- hours, minutes, seconds.

Function:- display(), addtime()

validation:- minutes and seconds should be valid

TollBooth class

Data:- totalcars, totalcash

Function:- payingCar(), nopayCar(), display()

validation:- Count cars and collect money.

Angle class

Data: degrees, minutes, direction

Functions:- getAngle(), display()

validation:- Direction must be N, S, E, or W. minutes must be less than 60.

Use of Const

```
void display() Const;
```

Const means the function will not change the object's data.

D:- Testing and validation

Test case	Input	Expected Result
valid circle	Radius = 5	Correct area
Valid Time	10:30:45 + 02:40:30	13:11:15
Invalid circle	Radius = -5	Invalid input
Boundary Time	10:59:59 + 00:00:01	11:00:00
Invalid Angle	minutes = 75	Invalid input

The End: